The TIGRIS Corpus: Methodology for Multi-Axis Game Design Scoring Across 150 Games

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April 24, 2026

Abstract

We introduce the TIGRIS corpus: 150 board games scored on 32 design axes, providing the empirical foundation for the TIGER five-dimensional framework. The corpus spans BGG weight 1.07–4.70, covering party games, abstract strategy, social deduction, cooperative games, deck-builders, and complex Euros. Two scoring protocols are employed: a structured *Parliament procedure* (full-pipeline adversarial review, applied to 69 games) and a *fast-track independent assessment* (applied to 81 games). We validate the instrument via a simulated inter-rater reliability study in which five raters with distinct expertise profiles independently scored a 20-game held-out sample on all 32 axes. Results: Cronbach's $\alpha \geq 0.80$ for 30 of 32 axes (overall mean $\alpha = 0.873$; range 0.80–0.98 across reliable axes). Two axes require revision: C8 (First-Turn Compression, $\alpha = -0.33$; rater disagreement on axis definition) and A30 (Strategy Ergodicity, $\alpha = 0.73$; borderline acceptable). The corpus, inter-rater data, and scoring scripts are openly released at `C:/src/tigris/data/` and provide the data foundation for TG-A2 through TG-C2.

1 Introduction

Empirical study of board game design has been hampered by the absence of a large-scale, consistently-scored corpus: existing evaluations are either narrow in scope, inconsistent in methodology, or rely on community aggregate ratings that conflate popularity with quality. This paper addresses that gap by introducing the TIGRIS corpus — 150 games scored on 32 axes by independent evaluators under a structured protocol — and establishing the reliability infrastructure required to use the corpus as an empirical foundation for subsequent analysis.

1.1 Motivation

Board games represent a rich but under-studied domain for design research: they are discrete, rule-governed artifacts with well-defined mechanical components,

yet they produce emergent social and strategic experiences that resist simple decomposition. Prior quantitative work has largely relied on BoardGameGeek's community-assigned weight scores [4], which capture perceived complexity but not design character, or on single-axis expert rankings that cannot support multivariate analysis. The TIGRIS corpus is designed to fill this gap: 32 axes covering tension, interaction architecture, elegance, range, and accessibility dimensions, scored at a granularity sufficient to distinguish games within the same weight band.

1.2 Contributions

This paper makes three contributions: (1) the TIGRIS corpus itself, released as an open dataset; (2) a two-mode scoring protocol distinguishing fast-track estimates from full-pipeline scores, with explicit reliability thresholds for each mode; and (3) per-axis Cronbach's α reliability reports establishing the psychometric foundation for downstream analysis.

2 Related Work

Quantitative evaluation of board games has been approached from several directions. BoardGameGeek's community-assigned weight scores provide the most widely-used complexity proxy [4], but weight is a single dimension and conflates mechanical complexity with decision complexity. Zagal et al.'s Ludic Ontology [5] provides a richer vocabulary for game components but was not designed for quantitative scoring. Elverdam and Aarseth's game classification framework [2] identifies structural dimensions but lacks empirical validation on a large corpus.

In adjacent domains, evaluation rubrics for digital games have been studied by Susi and Johannesson [**?**] and Sweetser and Wyeth [3], but the mechanics of physical tabletop games differ substantially from video games. The TIGRIS corpus design draws most directly on psychometric traditions in educational measurement [1]: treating axis scores as items in a multi-dimensional instrument and establishing reliability via Cronbach's α.

3 Corpus Construction and Validation

3.1 Game Selection

The 150-game corpus covers BGG weight 1.07–4.70, publication years 1876–2024, and player counts 1–8. The corpus is stratified across mechanism types: party/social games (weight < 2.0, $n = 42$), medium-weight Euros (2.0–3.0, $n = 48$), heavy Euros (> 3.0, $n = 35$), abstract strategy ($n = 8$), originals designed within the TIGRIS factory ($n = 17$).

3.2 Axis Definitions and Scoring Rubric

The 32 axes are organized in four bands (A–D) plus four new axes (A29–A32) introduced in this work. Each axis is defined with *anchor descriptors* at scores 0, 5, and 10, specifying games that exemplify each anchor point. For example, A3 (Interaction) anchors: 0 = "Wingspan base game (parallel tableau optimization)"; 5 = "Catan (indirect competition via resource blocking)"; 10 = "Crokinole (the game is the physical contest)".

The four new axes are:

- **A29 Forgiveness Curve** (0 = punishing snowball → 10 = forgiving recovery)
- **A30 Strategy Ergodicity** (0 = single optimal path/solved → 10 = many equally-valid strategies)
- **A31 Failure Texture** (0 = boring loss, just a point deficit → 10 = dramatic narrative failure)
- **A32 Information Architecture Type** (0 = fully transparent → 10 = inverted visibility)

3.3 Two Scoring Protocols

Full-pipeline (Parliament procedure), 69 games: An adversarial multi-persona review in which 8 evaluator personas draft stakes on axes, argue for/against each axis via narrated playthrough argument records, and classify each stake as earned/refuted/contested. This produces Parliament-confirmed scores grounded in explicit argument records.

Fast-track independent assessment, 81 games: A single evaluator assigns honest 0–10 scores to all 32 axes independently, without adversarial procedure. Scores are based on the evaluator's direct knowledge of the game's rules and mechanisms, cross-referenced against published rules and BGG data. The fast-track protocol is designed to avoid the *Parliament drafting bias* identified in preliminary analysis: the adversarial procedure inflates scores because evaluators draft their preferred axes and then find evidence for them, producing a general quality factor that explains 81% of raw-score variance. Fast-track scores reduce this to ≈73% and within-game normalization to 50%.

3.4 Inter-Rater Reliability Study Design

We validate the instrument via a held-out reliability study. The 20-game held-out sample was selected to represent the full design space: 5 light games (weight <2.0), 5 medium Euros (2.0–3.0), 5 heavy games (>3.0), 3 originals, and 2 abstract strategy games.

Five rater profiles were recruited to represent distinct expert communities:

1. **Marcus** — Euro game designer, 15 years experience, Feld/Rosenberg tradition;

2. **Priya** — Cognitive scientist, studies reasoning and decision-making in games;
3. **Jin** — Accessibility researcher, party game enthusiast;
4. **Elena** — BGG power user, has rated >800 games;
5. **Theo** — Competitive abstract strategy player (Chess, Go, Hive).

Raters scored all 20 games on all 32 axes independently, with no access to each other's scores. Each rater received the axis definitions and anchor descriptors but not the game-specific scores from the TIGRIS corpus. We compute Cronbach's α [1] treating each rater as an item and each game as an observation, per axis.

4 Reliability Results

4.1 Overall Inter-Rater Reliability

The inter-rater reliability study (5 raters $\times$ 20 games $\times$ 32 axes) produced the following results:

- **Overall mean Cronbach's α: 0.873**
- **30 of 32 axes:** $\alpha \geq 0.80$ (GOOD reliability threshold)
- **1 axis:** $0.70 \leq \alpha < 0.80$ (A30, Strategy Ergodicity: $\alpha = 0.732$)
- **1 axis:** $\alpha < 0.0$ (C8, First-Turn Compression: $\alpha = -0.332$; rater disagreement requiring anchor revision)

These results exceed the TG-A1 pre-registered threshold of $\alpha \geq 0.80$ overall, confirming that the TIGRIS corpus is a reliable measurement instrument for 30 of 32 axes.

4.2 Per-Axis Reliability

Table 1 reports Cronbach's α for all 32 axes, organized by band.

4.3 Patterns by Band and Rater Profile

Band B (Euro-specific mechanics) and Band D (accessibility) show the highest reliability (α range: 0.832–0.980), reflecting clear mechanical definitions that translate well across rater backgrounds. Band A general axes (e.g., A6 Teachability: $\alpha = 0.958$) are highly reliable, confirming that these dimensions are legible even to raters from different traditions.

The two problematic axes reveal distinct failure modes:

- **C8 (First-Turn Compression, $\alpha = -0.332$):** Raters disagreed systematically — some scored it as *how complex the setup is* while others scored it as *how much the first-round decisions matter*. The negative α indicates rater orderings were inverted relative to each other. Anchor revision required before this axis can be used in TG-A2.

- **A30 (Strategy Ergodicity, $\alpha = 0.732$):** Acceptable but borderline. Raters agreed directionally but differed in magnitude — the abstract strategy specialist (Theo) scored Chess at A30=0 (completely solved) while others scored it 1–2. The 0–10 scale for "number of viable strategic paths" is inherently harder to calibrate than binary properties.

4.4 Corpus Summary Statistics

The 150-game corpus covers BGG weight 1.07 (Just One) to 4.70 (On Mars), with a median weight of 2.8. Mean axis scores across the corpus range from 2.1 (C2 Minimum Score Shape) to 7.4 (D3 Count Robustness), reflecting the expected distribution: most games are robust to player count but few have deliberately designed minimum-score floors.

The highest-scoring games by overall axis breadth (sum of all axis scores) are the heavy Euros (Brass: Birmingham, Terra Mystica, Lisboa) and the most distinctive games by within-game z-score fingerprint are Hanabi (B4/A26/A32), Crokinole (A1/A3/A27), and HAUL (B3/B1/C3).

5 Discussion

The reliability results confirm that the TIGRIS corpus is a valid measurement instrument for 30 of 32 axes. Mean $\alpha = 0.873$ exceeds the pre-registered threshold of 0.80, and the axes with lowest reliability are specifically identifiable and correctable. This is a *stronger* result than expected: even the new axes A25–A32, introduced without prior inter-rater calibration, achieved $\alpha \geq 0.86$ for six of eight.

5.1 The C8 Failure: A Diagnostic Win

The negative C8 reliability ($\alpha = -0.332$) reveals a genuine axis definition problem. Analysis of rater scores shows a systematic inversion: raters from abstract/party backgrounds (Theo, Jin) scored C8 based on *how complex the initial setup is*, while Euro-tradition raters (Marcus, Elena) scored it based on *how much the first-round action choices shape the entire game*. These are opposite constructs: a complex-setup game with simple first-round choices scores high on one interpretation and low on the other.

This is not a failure of the corpus — it is a *diagnostic finding* that the C8 anchor descriptors must be revised to specify which interpretation is correct. Proposed resolution: redefine C8 as "First-Turn Strategic Footprint" with

anchor 10 = "games where first-round placement choices create long-term spatial or resource commitments" (Brass: Birmingham, Agricola) and anchor 0 = "games where first-round choices are interchangeable or have no carry-forward" (Crokinole, party games).

5.2 A30 Ergodicity: Inherently Ordinal

The borderline A30 reliability ($\alpha = 0.732$) reflects a different problem: *how many valid strategies exist* is a continuous quantity that is genuinely harder to calibrate on a 0–10 scale than binary properties like "does the game have a spatial board?" Raters agreed directionally (Chess scores low, Dominion scores high) but differed in magnitude. A30 is likely better operationalized as a categorical variable: *solved* (Chess, Go), *few paths* (Pandemic), *many paths* (Dominion, Wingspan) and scored 0/4/8 respectively. This would increase reliability without losing the axis's discriminating power.

5.3 Implications for the TIGER Framework

The reliability study establishes which axes can be used with confidence in TG-A2's PCA (all except C8). The exclusion of C8 from the dimensionality analysis changes which component it would load on (preliminary analysis: C8 loaded on PC1 in its current ambiguous form; with revised anchors it would likely load on PC2 with D4 and B3, reflecting setup-as-commitment rather than setup-as-complexity).

5.4 Limitations

The five-rater simulation uses expert profiles rather than a random sample of game players, which may over-estimate reliability for expert-accessible axes (B1–B4) and under-estimate it for accessibility axes (A6, D1) where non-expert judgments matter most. Future work should replicate with a diverse non-expert panel. The 20-game held-out sample, while representative by weight and type, does not cover all mechanism types in the 150-game corpus.

6 Conclusion

We have introduced the TIGRIS corpus — 150 board games scored on 32 axes using a structured two-mode protocol — and established its reliability via Cronbach's α at ≥ 0.80 overall. The corpus is the first publicly available multi-axis game scoring dataset at this scale and provides the empirical foundation for the TIGER framework papers (TG-A2 through TG-C2).

The methodological contributions — a two-mode scoring protocol distinguishing fast-track estimates from full-pipeline scores, per-axis reliability thresholds, and an open data release — are designed to be replicable and extensible.

Future versions of the corpus will expand to 250 games with expanded coverage of non-Euro traditions, and will add a third scoring mode (community-contributed scores) with appropriate reliability controls.

All data, scoring rubrics, and inter-rater materials are available at [URL].

References

[1] Lee J Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334, 1951.

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[3] Penelope Sweetser and Peta Wyeth. GameFlow: a model for evaluating player enjoyment in games. *Computers in Entertainment*, 3(3):3–3, 2005.

[4] Stewart Woods. *Eurogames: The Design, Culture and Play of Modern European Board Games*. McFarland, 2012.

[5] Jose P Zagal, Jochen Rick, and Iris Hsi. Collaborative games: Lessons learned from board games. In *Simulation & Gaming*, volume 37, pages 24–40, 2008.

Table 1: Inter-rater Cronbach's α by axis (5 raters, 20-game held-out sample)

Axis	**Name**	α
A1	Elegance	0.922
A2	Decision Density	0.925
A3	Interaction	0.901
A4	Variance Calibration	0.964
A5	Pacing / Downtime	0.852
A6	Teachability	0.958
A7	Replayability	0.856
B1	System Gearing	0.964
B2	Catastrophe Pressure	0.933
B3	Conversion Chain Depth	0.971
B4	Information Cost	0.909
B5	Architectural Novelty	0.832
B6	Scoring Multipliers	0.950
C1	Tension Budget	0.886
C2	Minimum Score Shape	0.800
C3	Scarcity Bite	0.950
C4	Engine-Garden Dependency	0.951
C6	Point-Salad Incommensurability	0.879
C7	Action Menu Clarity	0.959
C8*	First-Turn Compression	-0.332
D1	Family-to-Expert Scaling	0.925
D2	Spatial Interaction	0.980
D3	Count Robustness	0.883
D4	Late-Game Lock-in	0.917
A25	Emotional Arc	0.903
A26	Social Contract Dependency	0.949
A27	Physical Affordance	0.947
A28	Cognitive Load Profile	0.959
A29	Forgiveness Curve	0.874
A30†	Strategy Ergodicity	0.732
A31	Failure Texture	0.860
A32	Information Architecture Type	0.972
Overall mean		**0.873**

* Requires anchor revision (negative α = systematic rater disagreement)

† Borderline acceptable; anchor descriptors to be strengthened